Which stored elastic tensors in JARVIS-DFT are not positive definite? An exact-rational Born-stability audit of 34 323 tensors

(a statement about stored DFT tensors, not about materials or about JARVIS being wrong)

Xavier Callens (SocrateAI Lab)

Preprint v1, 2026-10-31

doi:10.5281/zenodo.23842486 (not peer reviewed)

Zenodo DOI to be assigned on release (not peer reviewed)

AI-assistance disclosure and audit status. This preprint, its Python code, its Lean code and its claim ledger were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was independently audited by a different AI agent instance (`AUDIT.md`, `ACCEPT_WITH_CORRECTIONS`), not by a human.** The audit's corrections C1–C10 are applied in this text (Section 7). No human audit and no statement lock exist, and the binding `lean_runner.py` gate was not run; these are owed (Section 9).

Abstract

An unstressed crystal is elastically stable only if its 6×6 stiffness matrix C is positive definite (the Born criterion, as stated by Mouhat and Coudert [1]). We tested every full 6×6 elastic tensor stored in the JARVIS-DFT `dft_3d` dataset [2] (CC BY 4.0; figshare article 6815699; unzipped json sha256 `9dacb55a...5907`) with exact rational arithmetic. Of 93 902 records, 59 400 carry no tensor (`"na"`), 179 contain a non-finite number, and $N = 34\,323$ are analysed. Reading each stored float as the exact rational of its shortest decimal, and testing the symmetric part $S = (C + C^T)/2$: 22 842 tensors are positive definite and 11 481 **(**33.45%**) are not**, each with an exact rational witness $x \neq 0$, $x^T S x \leq 0$. Three exact methods agree on every entry (the producer's LDL^T and Bareiss elimination, and an independent auditor's characteristic-polynomial sign pattern), and the auditor re-checked all 11 481 witnesses. 234 tensors are numerically zero; without them 11 247 of 34 089 (32.99%) are not positive definite. Under an exact worst-case rounding test (each printed value within half a unit of its last printed place) 22 809 verdicts are robustly positive definite, 11 244 robustly not, 55 certified ambiguous and 215 undecided (210 of them null tensors). 342 tensors pass the incorrect linear orthorhombic criterion but are not positive definite. **This is a statement about stored DFT tensors under one stated reading.** Many entries are raw DFT outputs, including hypothetical, unrelaxed and dynamically unstable structures, so a large fraction is expected. It does **not** say that JARVIS-DFT is wrong and says nothing about any material being unstable in a laboratory. The kernel-checked Lean part (Tier [A], provisional: a statement-lock entry and the `lean_runner.py` gate are still owed) covers 18 explicit matrices and 2 control statements only.

What this is NOT. Not a claim that any material is stable, unstable or synthesisable; not a claim that JARVIS-DFT or its authors are wrong; not a formalisation of the classification (Lean covers 18 matrices and 2 controls); not a statement under applied load (the zero-stress criterion is tested). The owner has not forwarded the dataset observations of Section 3.6 to the dataset authors.

Markers. [A]: kernel-checked Lean, axioms within {`propext`, `Classical.choice`, `Quot.sound`}, controls caught (a statement-lock entry and the `lean_runner.py` gate are still owed, so [A] is provisional). [L]: a cited source, taken as stated. [C]: executed code with an argument written down but not machine-checked (here: exact rational arithmetic on the stored decimals). [X]: float runs, context counts and recorded absences, supporting only.

1 Setting and question

Criterion (Tier [L]). Mouhat and Coudert [1] state that an unstressed crystal is elastically stable iff the quadratic form $E = \frac{1}{2} V_0 \sum C_{ij} \varepsilon_i \varepsilon_j$ is positive for all $\varepsilon \neq 0$, i.e. iff the 6×6 matrix C is positive definite (all eigenvalues > 0, equivalently all leading principal minors > 0). They cite Born and Huang [4] for this; we did not read that book. Under load the relevant matrix is $B \neq C$ (their Eq. 20), so what is tested here is the *zero-stress* criterion (correction C4). Their Eq. 19 is the linear orthorhombic criterion of Wu et al., which they call incorrect, with the counterexample of their footnote 26. Only the symmetric part $S = (C + C^T)/2$ enters $x^T C x$, so verdicts are for S; the number of non-symmetric stored tensors is reported separately.

Data and licence (Tier [L]). JARVIS-DFT `dft_3d` [2, 3], figshare article 6815699 (DOI `10.6084/m9.figshare.6815699`; the file used is `jdft_3d-9-24-2025.json.zip`, figshare version 11). The unzipped json has sha256 `9dacb55ac8371c7c2bf0332933b55c98e9cca6201e92eb9d6fcd03aab0975907`; the zip's md5 `c3179161...` equals the figshare API's `computed_md5`. The licence is **CC BY 4.0**, read from the figshare API by the producer before download and again by the auditor; attribution: Choudhary et al., npj Comput. Mater. 6, 173 (2020). The dataset is not redistributed with this paper (no bulk data, no per-entry table; per-entry results are kept outside the repository, with their sha256 in `data/result.json`). Exactly 22 dataset tensors are shipped: 18 in `Witnesses.lean` and 4 in `data/summaries/f1_inspect.json`, each file carrying the CC BY 4.0 attribution line (correction C9). Stored units are GPa (the auditor matched five published values of Table S3 of [3] to the printed digits).

Questions (preregistered). `data/prereg_M2.json` (sha256 `2c3127f9...5d0fb`) asked: (Q1) how many tensors are certifiably not positive definite when the stored decimals are read as exact rationals; (Q2) how many verdicts are ambiguous under half-unit rounding; (Q3) how many pass the incorrect linear criterion but fail exact positive definiteness; (Q4) where float pipelines disagree with the exact verdict; (Q5) where the class-specific closed-form conditions of [1] disagree with the exact verdict. The outcome rule was KEEP if all controls pass, the whole dataset is classified, two implementations agree and the certified not-positive-definite fraction is at least 1%. Outcome: KEEP.

2 Method

Reading. Each stored float is converted to the exact rational of its shortest round-trip decimal representation. This is an interpretation: the stored decimals are treated as exact numbers. A DFT tensor is a calculation, not "the material's" tensor. The auditor checked that this equals reading the literal json text for every finite entry (1 235 628 entries, 0 differences) and that no entry is a json integer literal.

Exact test. [C] LDL^T without pivoting over $\mathbb{Q}$ on S: S is positive definite iff all pivots are > 0 (Sylvester). If the first non-positive pivot is $d_k \leq 0$, solving $L^T x = e_k$ gives an exact rational witness with $x^T S x = d_k \leq 0$ (a "boundary" case when $d_k = 0$). The producer ran this (LDL^T), an integer Bareiss elimination on all entries, and sympy leading minors on a 14 615-entry subset (all not-positive-definite entries, all rounding-undecided entries and 3 000 random positive definite ones; a logged deviation from "all"). The auditor wrote separate code deciding positive definiteness from strict sign alternation of the exact characteristic polynomial over $\mathbb{Q}$ (an "iff" because S is symmetric, hence real-rooted) on all 34 323 entries.

Ordering. JARVIS stores the VASP order (xx, yy, zz, xy, yz, zx). Positive definiteness is invariant under simultaneous permutation and positive diagonal congruence, so the order cannot change a full-tensor verdict; it matters only for the class conditions, where the producer maps to Voigt by $[0, 1, 2, 4, 5, 3]$. The auditor confirmed the mapping (tetragonal-I entries: stored index 4 equals index 5 exactly in 9 596 of 10 343, against index 3 equals index 4 in 35; the identity map would give 1 046 instead of 555 class disagreements).

Rounding robustness. [C] Each printed value is allowed $\pm\frac{1}{2}$ unit of its last printed place (half-widths Δ_{ij}, symmetrised). The producer used sufficient tests (a Frobenius-norm bound for "robustly positive definite", $x^T S x + |x|^T \Delta |x| < 0$ for "robustly not"). The auditor replaced the first by the exact Rohn vertex test: every

symmetric $S + E$ with $|E_{ij}| \le \Delta_{ij}$ is positive definite iff all 32 matrices $S - T_z \Delta T_z$, $z \in \{\pm 1\}^6$, $z_1 = 1$, are (here $T_z = \mathrm{diag}(z)$). A not-positive-definite entry is "certified ambiguous" when some $S + T_z \Delta T_z$ or $S + \mathrm{diag}\,\Delta$ is positive definite; "undecided" means no certificate was found.

Float pipelines (Tier [X]). `numpy.linalg.eigvalsh` on S with three conventions: F0 $\lambda_{\min} > 0$; F1 $\lambda_{\min} > -10^{-8}$; F2 $\lambda_{\min} > -10^{-6}\lambda_{\max}$. These are context, not evidence for the exact verdict.

Code. `data/code/`: `born_exact.py` (exact tests, class conditions, intervals), `controls.py`, `run_dataset.py`, `analysis2.py`, `final_summary.py`, `inspect_f1.py`, `gen_lean.py`, `make_result_json.py`. The downloaded json was only parsed with `json.load`; the producer's own scripts were not run with `-I` (it would drop the script directory), a logged deviation.

3 Results

3.1 Headline counts (Tier [C])

quantity	count
records in the dataset	93 902
with `elastic_tensor = "na"`	59 400
with a 6×6 list	34 502
of which with NaN or $\pm\infty$ (excluded)	179
analysed tensors N	34 323
exactly positive definite	22 842
exactly not positive definite (each with a verified rational witness)	**11 481** (33.45%)
of which on the boundary (first non-positive pivot exactly 0)	204
numerically null tensors (max $\lvert\text{entry}\rvert < 10^{-6}$; all not positive definite)	234
not positive definite without the null tensors	11 247 of 34 089 (32.99%)
LDL^T vs Bareiss disagreements	0
auditor (characteristic polynomial) vs producer, 10 fields, all 34 323 entries	0 mismatches

The 234 null tensors have entries from 10^{-14} down to $5 \cdot 10^{-324}$; the largest maximum entry below the cut is $6.8 \cdot 10^{-12}$ and the smallest above is 0.333, so the 10^{-6} cut is not arbitrary in effect. The auditor re-checked all 11 481 witnesses on its own S ($x \neq 0$, $x^T S x \le 0$, equal to the stored value), superseding the requested 500-sample.

3.2 Rounding robustness (correction C1)

half-width rule	robust PD	robust not PD	certified ambiguous	undecided
$\pm\frac{1}{2}$ unit, producer (Frobenius)	22 708	11 244	—	371
$\pm\frac{1}{2}$ unit, auditor (exact Rohn)	**22 809**	11 244	55 (33 PD, 22 not PD)	215 (210 null)
floored at 0.05, producer	22 708	11 216	—	399
floored at 0.05, auditor (Rohn)	**22 809**	11 216	286 (33 PD, 253 not PD)	12

33 399 tensors print every entry at one decimal; 874 have an entry printed with more than six places (numerical noise, down to denormal numbers), where the $\pm\frac{1}{2}$-unit rule gives absurdly small intervals; this is why the floored variant was added post hoc. The producer's bounds are sufficient, not necessary, so its "undecided" means "not certified"; the auditor's Rohn test closes this: of the 134 producer "positive definite, undecided" entries, 101 are robustly positive definite and 33 truly ambiguous, and every producer-robust verdict is also auditor-robust. Excluding null tensors (at $\pm\frac{1}{2}$ unit): robust PD 22 809, robust not PD 11 225, ambiguous 50, undecided 5.

3.3 The incorrect linear criterion and the class conditions

Q3. 23 184 tensors pass the linear orthorhombic criterion of Wu et al. [5] ([1] Eq. 19) applied to all entries regardless of class; 342 (1.00% of N) of them are exactly not positive definite (328 robustly); 0 fail it while positive definite, consistent with necessity. By class, pass-but-not-PD: cubic 87, tetragonal I 63, trigonal I 57, monoclinic 47, triclinic 41, hexagonal 29, orthorhombic 15, trigonal II 3, tetragonal II 0.

Q5 (correction C3). Closed-form class conditions (cubic, hexagonal, tetragonal I/II, trigonal I/II, orthorhombic; Eqs. 6, 9, 11, 13, 15, 18 of [1]) on the stored entries of 30 203 tensors disagree with exact positive definiteness of the full tensor in 555 cases: 544 class-stable but not positive definite, 11 class-unstable but positive definite. Most are expected: the class formulae read only representative entries (e.g. C_{11} for C_{22} and C_{33}, C_{44} for C_{55} and C_{66}), and the stored tensors do not satisfy those class equalities exactly. The auditor's split of the 544 by block: 184 have a Voigt diagonal entry ≤ 0, 299 a non-positive-definite normal 3×3 block, 55 a non-positive-definite shear block, and 6 fail only through normal–shear couplings. So 483 are driven by violated class equalities and 61 by off-diagonal couplings (mostly outside the class zero pattern). This is not an error in either method. Monoclinic and triclinic reduce to the generic test and are not cross-checked.

3.4 Float pipelines (Tier [X])

F0 disagrees with the exact verdict on 0 tensors. F1 (-10^{-8} absolute) disagrees on 236, all calling a not-positive-definite tensor stable; 234 are the null tensors and 2 are not. F2 (relative $-10^{-6}\lambda_{\max}$) disagrees on 3 (2 not null). The disagreements are essentially tolerance verdicts on null-like tensors plus two near-singular tensors. The auditor's own numpy call reproduces the figures.

3.5 DFT-uncertainty context (illustrative)

Exact test of $S - \theta\|S\|_F I$ among the 22 842 positive definite tensors: $\theta = 1\%$: 22 074; 2%: 21 224; 5%: 17 983; 10%: 10 703; 20%: 611. The dataset stores no uncertainty; these thresholds are illustrative stand-ins and not statements about JARVIS accuracy.

3.6 Dataset observations (neutral)

- 59 400 of 93 902 records have `elastic_tensor = "na"`; 179 tensors contain NaN or infinity.
- 234 stored tensors are numerically zero; 12 tensors have a literal zero row.
- 922 tensors are not exactly symmetric: 232 of them are null tensors, 598 have $\max|C_{ij} - C_{ji}| > 1$ GPa (median 19.5 GPa), and for 7 the verdict depends on which triangle is mirrored (correction C5).
- 874 tensors have an entry printed with more than six decimal places; 18 tensors have an entry above 10^4 GPa; the smallest float eigenvalue reaches $-6.3 \cdot 10^5$.
- Read-sanity (a positive control for reading the dataset): Voigt K and G recomputed from the tensor match the stored `bulk_modulus_kv` and `shear_modulus_gv` within 1% for 97.9% and 98.1% of entries (median relative error $3 \cdot 10^{-5}$).

The owner has not forwarded these observations to the dataset authors; whether to do so is the owner's decision.

3.7 Context counts (Tier [X], no association claimed)

Among entries with `min_ir_mode`, a negative lowest IR mode is stored for 40 of 480 not-positive-definite tensors (8.3%) against 221 of 2 959 positive definite ones (7.5%); the meaning of the field was not verified. The not-positive-definite fraction is 11.3% (1 048 of 9 258) for entries with an ICSD id against 41.1% (10 199 of 24 831) without. **The `ehull` bins of the producer's report are withdrawn** (correction C2): the field was not verified to be an energy above hull (median 1.55, ICSD-linked median 1.46, spot checks mixed), so it is not used here; the shipped producer json files still contain those bins and should be ignored. A large fraction of the dataset is hypothetical structures, and we do not know the convergence state of individual calculations.

4 Interpretation

The percentage is the fraction of stored DFT tensors that are not positive definite under one stated reading (exact decimals, symmetrised, zero-stress criterion). A tensor with a negative eigenvalue is a calculated tensor that is not positive definite; finite-difference noise, unrelaxed or hypothetical structures, residual stress, magnetism and convergence all remain possible reasons, and finite-difference tensors of dynamically unstable cells are expected to be non-positive-definite at times. The criterion tested is the zero-stress Born criterion (Mouhat–Coudert Eq. 20 replaces C by B under load); residual stress in a computed cell changes the relevant matrix. The JARVIS elastic-property paper [3] invites users to apply Born's criterion but, to the extent the auditor read it, reports no unstable fraction. **Novelty:** to our knowledge no exact-rational Born audit of the whole JARVIS elastic dataset has been published; this is not an exhaustive search and not a priority claim. The result neither validates nor invalidates any individual entry or the dataset.

5 Controls

All run and logged before any dataset number (`data/logs/controls.json`, the tracked record; `controls.log.txt` is a copy of the untracked `controls.log`: 24 lines, every check PASS, `ALL_CONTROLS_PASS True`; correction C8).

Positive. Cu, Si and diamond cubic tensors (literature values) are positive definite in exact, Bareiss and float arithmetic, the cubic Born conditions agree and the interval verdict is robust PD. The Mouhat–Coudert footnote-26 tensor ($C_{11} = \frac{1}{2}$, $C_{22} = C_{33} = 1$, $C_{12} = -1$, $C_{13} = C_{23} = 0$; the shear constants are not given in the footnote and are set to 1 here, labelled) passes the incorrect linear criterion, fails exact positive definiteness and the orthorhombic Born conditions, has a rational witness and is robustly not positive definite. Three known-unstable cubic tensors are not positive definite. The auditor additionally reproduced stored JVASP-91, 1002, 890, 1174 and 1195 against the printed Table S3 values. Read-sanity of Voigt K, G against the stored moduli (Section 3.6).

Negative. Sign flips of C_{11} and C_{44} on each textbook crystal flip the verdict; symmetry breaking is detected and a large one flips the verdict; a singular tensor ($C_{11} = C_{12}$) is not positive definite and rounding-undecided; a tensor 0.1 GPa from the boundary is positive definite but undecided. Random cross-check: exact LDL^T against integer Bareiss on 6 000 random rational matrices agree, plus sympy minors on 400.

Auditor's controls. Fourteen of its own, all PASS: diamond, Cu positive definite; $C_{11} < C_{12}$, $C_{44} < 0$, $C_{11} = C_{12}$ and footnote 26 not positive definite; an exact determinant check; characteristic polynomial against elimination against numpy on 3 000 random symmetric rational matrices; Rohn robust, ambiguous and near-boundary cases; the Mouhat–Coudert cubic conditions. The auditor did not re-run the producer's `controls.py`.

6 Lean (Tier [A], provisional)

Files in `formal_cert/Programmes/BornM2/` (copies in `data/lean/`), `import Mathlib.Tactic` (never `import Mathlib`), compiled with `scripts/certified_numerics/lean_pinned.py` (Lean 4.34.1, Mathlib `d13f23b7`).

- `Probe.lean`: `foot_witness` (a witness $x \neq 0$ with $x^T C x \leq 0$ for the footnote-26 tensor) and `dia_pd` ($578 \sum x_i^2 \leq x^T C x$ for a diamond-type tensor, which does imply positive definiteness in Lean terms). Axioms: `propext`, `Classical.choice`, `Quot.sound` only.
- `Witnesses.lean`: 12 theorems `notpd_k` $\exists x \neq 0,\ x^T S x \leq 0$ for symmetrised stored tensors of 12 dataset entries (4 pass the linear criterion; cubic 2, hexagonal 2, tetragonal I 2, orthorhombic 1, monoclinic 1, others as in `code/gen_lean.py`, a deterministic sample), and 6 pairs `pd_k`, `pd_k_pivots` for 6 randomly chosen robustly positive definite entries: the identity $x^T S x = \sum_k d_k (l_k \cdot x)^2$ with explicit rational d_k and L unit lower-triangular, and $0 < d_k$. **Scope note (correction C6):** `pd_k` proves the factorisation identity and the positive pivots; positive definiteness follows from L being unit lower-triangular, which is visible in the statement, but that step is not a Lean theorem.

- `Tamper1/2/3.lean` (wrong witness; lower bound $600 > 578$; altered pivot) are **rejected** by the kernel, rc 1 (errors: unsolved goal $2 \leq 0$, `linarith` failure, `ring` failure). These are controls.

After the CC BY 4.0 attribution comment was added to `Witnesses.lean` (correction C9) it was recompiled with `lean_pinned.py`: rc 0, 24 `#print axioms` lines, each within the whitelist, no `sorryAx` (`data/logs/lean_witnesses_after_c9.json`; the pre-edit transcript `lean_witnesses.json` is kept). The auditor compiled byte-identical copies under its own module names (Probe rc 0, Witnesses rc 0, Tamper1–3 rc 1), replayed both modules with `leanchecker` (rc 0; a missing-module negative control rc 1), found no escape hatch in a grep over all five files, and matched all 18 matrices and 12 witnesses to its own symmetrised tensors by jid (18 of 18 identical; 12 of 12 witnesses give $x^T S x < 0$; the 6 `pd_k` matrices are auditor-positive definite). Those were checks on the *pre-attribution* bytes; the only change since is the added comment.

Labelling. Kernel-checked and independently replayed, but **not yet Tier [A] in the strict sense**: that needs a statement-lock entry and a run of the `anse/formal/lean_runner.py` gate (compile, `#print axioms`, `sorryAx` rejected), neither of which has been done; both are owed. The Lean files state facts about the exact matrices written in them (symmetrised stored decimals); matching them to the dataset is a Python and hash step (Tier [C]). **Not in Lean:** the classification of the other 11 469 unstable entries and of the stable ones (verified exactly in Python only), the rounding tests, and the link between the criterion and crystal stability.

- `Probe.lean`: `bfd6b8919ae197d63d4bb04a5dbe88b6a0a6ed0f46b6bffe4c0351642e270c90`
- `Witnesses.lean`: `c134b1390e058efb3af1b8ee2532f54a2a44f7a79d502d5d4981d970387de427` (with attribution comment; pre-edit `1c83a5a0...9bafc`)
- `Tamper1.lean`: `43b4050236e46967327038b00442e47578ee698ccb38b550fbcf3e863df9f488`
- `Tamper2.lean`: `d17051b84043c67796eaae7f26a8e14b0dd5583a993887c2266bef9f6a6f7142`
- `Tamper3.lean`: `0159b8ee0c1646e87bc88f9e7d0070bf3ff8db53e330ddf5f31f85c6ed0cff18`

7 Audit corrections applied

The independent audit (`AUDIT.md`, §9) listed ten corrections. Their treatment:

C1 Exact Rohn rounding numbers replace the sufficient-test numbers (Section 3); also mirrored in `data/result.json` under `interval_exact_rohn`.

C2 The `ehull` bins are withdrawn (Section 3); the field is kept neutral.

C3 The class-condition explanation and the 184/299/55/6 split are used (Section 3).

C4 The zero-stress caveat (Mouhat–Coudert Eq. 20) is stated in Sections 1 and 4.

C5 The size of the asymmetry (232 null; 598 above 1 GPa, median 19.5 GPa) is given (Section 3.6).

C6 The Lean heading and tier are relabelled and the `pd_k` scope note added (Section 6); `result.json`'s `tiers.lean_sample_and_controls` is updated accordingly.

C7 `prereg_M2.json` records `written_utc` 16:30Z, which is wrong: its file mtime is 16:22:52Z, before the dataset download (16:23:00Z) and all runs (16:24 to 16:37). The order is fine; the file was committed together with the results (commit `051f87d`), so git does not show it and only the mtimes do. The prereg file is not edited.

C8 The `logs/*.log` files are untracked (ignored by `*.log`); this paper cites `logs/controls.json` (tracked) and ships the logs in the data copy as `*.log.txt` (the bundler keeps only listed suffixes).

C9 The CC BY 4.0 attribution line was added to `Witnesses.lean` (as a comment) and to `data/summaries/f1_inspect.json` (as an `_attribution` key); the hashes in `data/result.json` were updated and `Witnesses.lean` recompiled, rc 0 (Section 6). The producer's `RESULT.md` (pre-audit text, hash unchanged) is superseded by this paper on C1–C10.

C10 The audit has been done by a separate AI agent instance (not a person); the `lean_runner.py` gate and a statement lock remain owed. The producer's "what an independent auditor must redo" list is thereby discharged except for those two items.

8 Claim ledger

`ledger/build_ledger.py` files 16 claims: 2 at Tier [A] (`BM-A-0001` the 18 matrices, `-0002` the probe and tamper controls), 3 at [L] (`BM-L-0001..0003`: the Born criterion, the dataset and licence, Born–Huang unread), 8 at [C] (`BM-C-0001..0008`, the last being the headline) and 3 at [X] (`BM-X-0001..0003`: float pipelines, context counts with `ehull` withdrawn, and what was not done). Each carries a content-addressed evidence blob in `ledger/evidence/`; each Tier [A] claim also carries an `ai_audit` field. The gate field `audit` is left null: the audit was done by an AI agent instance, not by a person. The Elenchus ledger gate (`ledger/run_gate.py`, `ledger/gate_output.txt`) therefore exits with code 1: 0 blocking findings and 2 flags `LEDGER_UNAUDITED_TIER_A`, one per Tier [A] row. This is the honest state.

9 Limitations

- **AI audit only; human audit and statement lock owed.** No person has certified that the Lean statements mean what their gloss says; the `lean_runner.py` gate was not run by producer or auditor.
- **Unverified, from the audit.** (a) The physical meaning of the JARVIS fields `ehull` and `min_ir_mode`, and the convergence state of individual calculations. (b) Born and Huang (1954) was not read; the criterion is taken from Mouhat–Coudert. (c) The producer's own `controls.py` was not re-run by the auditor, which ran its own controls. (d) The auditor's code and per-entry output are scratch files not shipped here; its reproduction is documented in `AUDIT.md` but cannot be re-run from this bundle. (e) No Lean certificates exist beyond the 18 entries plus 2 controls.
- **Reading.** Everything depends on taking each stored float as the exact rational of its shortest decimal and on testing the symmetric part. A different reading (e.g. the tensor as the rounded value of an unknown exact tensor) is what the rounding-robustness analysis addresses, with the limits given there.
- **Zero-stress criterion.** Not valid as a stability statement for a cell under residual stress.
- **Deviations from the preregistration** (logged in `data/RESULT.md`): sympy on a 14 615-entry subset (Bareiss on all, and the auditor's characteristic polynomial on all, close the gap); post hoc additions (half-width floor, null-tensor exclusion, K/G sanity, context counts, asymmetry test); two float tolerance conventions fixed in the prereg and used as written; Lean limited to controls and an 18-entry sample (the prereg called it a stretch); producer scripts not run with `-I`; the wrong internal timestamp of the prereg (C7).
- **Not run:** the repository guards `antigravity_guard.py`, `test_rigor_guard.py` and `pytest tests/` were not run for this paper.
- **Scope.** Only stored 6x6 tensors of one dataset version (9-24-2025). No claim about any material, any laboratory, any other database, or JARVIS's accuracy.

10 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: `abe4718ed552e3c77b7207cadb8095d998e02602`; the M2 audit file, the attribution edits and this paper are uncommitted working-tree files at that time. Copies of the inputs are in `data/` (`code/`, `lean/`, `logs/`, `summaries/`, `RESULT.md`, `result.json`, `prereg_M2.json`, `AUDIT.md`). `summaries/` is the lane's `data/` directory. The hashes in `result.json` are keyed by the lane's paths (e.g. `data/summary.json`); the log files there are `*.log` (here `*.log.txt`). The dataset itself is not shipped.

```
# fetch jdft_3d-9-24-2025.json.zip from figshare article 6815699 (CC BY 4.0), check sha256 of the unzipped json
# 9dacb55ac8371c7c2bf0332933b55c98e9cca6201e92eb9d6fcd03aab0975907
python code/controls.py ; python code/run_dataset.py ; python code/analysis2.py ; python code/final_summary.py
python scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/BornM2/Witnesses.lean
python -I papers/p4_born_stability_jarvis/ledger/build_ledger.py ; python -I papers/p4_born_stability_jarvis/ledger/run_gate.
    py
```

References

[1] F. Mouhat, F.-X. Coudert, *Necessary and sufficient elastic stability conditions in various crystal systems*, Phys. Rev. B 90, 224104 (2014), arXiv:1410.0065. The criterion, Eqs. 6, 9, 11, 13, 15, 18–20 and footnote 26 ([L]).

[2] K. Choudhary et al., *The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design*, npj Comput. Mater. 6, 173 (2020), arXiv:2007.01831; data: figshare article 6815699, doi:10.6084/m9.figshare.6815699, CC BY 4.0 ([L]).

[3] K. Choudhary, G. Cheon, E. Reed, F. Tavazza, *Elastic properties of bulk and low-dimensional materials using van der Waals density functional*, Phys. Rev. B 98, 014107 (2018), arXiv:1804.01033; read by the auditor for Table S3 and its remarks on Born's criterion.

[4] M. Born, K. Huang, *Dynamical Theory of Crystal Lattices*, Oxford (1954); cited second-hand through [1], not read.

[5] Wu et al., Phys. Rev. B 76, 054115 (2007); the source of the linear orthorhombic criterion (as cited in [1]; not read).

[6] The mathlib Community, *The Lean mathematical library*, arXiv:1910.09336 (2019); cited for the library only.